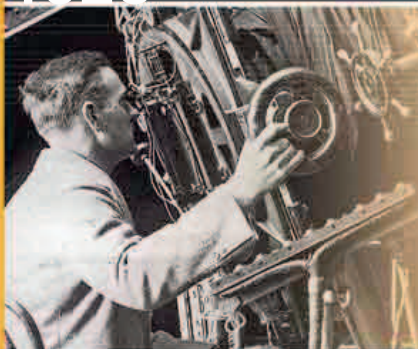


“WE CAN... CONSIDER THAT **FIRST** PIECE OF ORGANIC **SLIME...** AS BEING THE FIRST ORGANISM.”

A. Oparin, from *Origin of Life*, 1924

RUSSIAN BIOCHEMIST ALEXANDER OPARIN (1894–1980) proposed a theory regarding the **origins of life**. He claimed that the first **living things had evolved from nonliving matter**. He also suggested that Earth's earliest atmosphere was highly reduced—that is, gases had an abundance of bonded hydrogen atoms, just like molecules in organisms. The diversity of these gases provided the other elements needed as building blocks for these molecules: methane as a source of carbon, ammonia for nitrogen, and water vapor for oxygen. Oparin suggested that the gases reacted to form a “soup” of organic molecules, from which the first living cells evolved. In 1953, American chemist Stanley Miller (1930–2007) would show that organic molecules could be made from nonorganic ones.

Russian physicist Alexander Friedmann (1888–1925) was studying the curvature of space. One of his models suggested that the **radius of the universe** was constantly **increasing** with time. The idea of an expanding universe would be independently proposed by Belgian astronomer Georges Lemaître (see 1927); and a few years later, the American astronomer Edwin Hubble would collect evidence to suggest that galaxies were indeed receding (see 1929–30).



Edwin Hubble observes the stars through the enormous Hooker Telescope in California.

IN OCTOBER, EDWIN HUBBLE suggested that certain **stars of the Andromeda nebula** were much **farther away** than previously supposed—even beyond the Milky Way. His findings were revolutionary, because at the time scientists thought that the Milky Way was the extent of the entire universe.

At about the same time, American chemists **Gilbert**

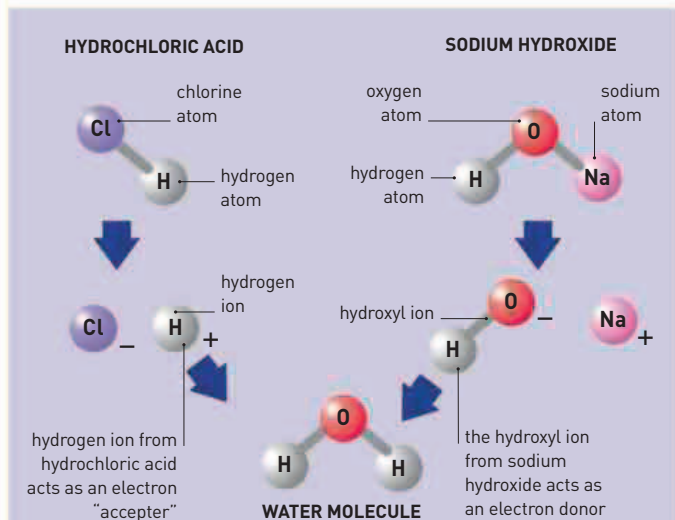
Lewis (1875–1946) and **Merle Randall** (1888–1950) published *Thermodynamics and the Free Energy of Chemical Substances*, which explained **chemical reactions in terms of energy**. In the 19th century, American physicist Willard Gibbs (1839–1903) had shown how the energy associated with chemical reactions could be quantified. Different substances have varied

“THE HISTORY OF **ASTRONOMY** IS A HISTORY OF **RECEDING HORIZONS.**”

Edwin Hubble, American astronomer, from *The Realm of the Nebulae*, 1936

amounts of chemical energy associated with them depending on their chemical composition; this affects their tendency to react. When a chemical reaction occurs, substances change from one form to another—and so does their chemical energy. According to the **laws of thermodynamics**, the total energy stays the same. So if products have less chemical energy than reactants, then some energy, such as heat, must be released. In their book, Lewis and Merle introduced “free energy” values for different substances, which would enable scientists to calculate energy changes associated with reactions.

Chemists also redefined acids and bases. In 1884, 19th-century Swedish chemist **Svante Arrhenius** had shown that **acids and bases** could be recognized by a prevalence of hydrogen (H^+) and hydroxyl (OH^-) ions respectively. Now, Danish chemist **Johannes Brønsted** (1879–1947) and British chemist **Martin Lowry** (1874–1936) independently defined them in terms of hydrogen ions alone: an acid donates hydrogen ions and a base accepts them. Gilbert Lewis took this one step further and explained them in terms of electrons—the negative subatomic particles that determine the chemical properties of all substances.

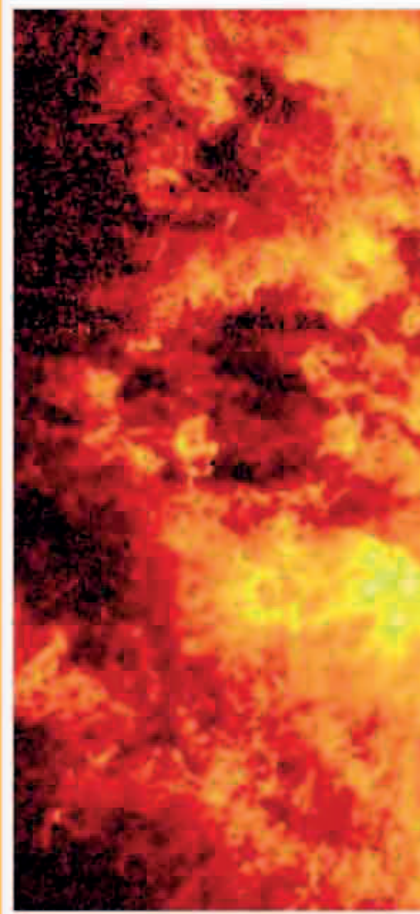


ACID-BASE THEORY

This theory defines acids and bases in terms of electrons. Acids are electron acceptors: that is, they release positively charged hydrogen ions (protons), which readily combine with negative electrons. Bases are electron donors—they provide the electrons that the hydrogen ions accept. Many bases—called alkalis—release electron-rich hydroxyl ions, which then react with hydrogen ions to form water.



THE YEAR BEGAN with another important development in the field of astronomy. In 1920, British astronomer **Arthur Eddington** had proposed that a star's energy came from the nuclear fusion of hydrogen to form helium, a theory later supported by fellow British



May Alexander Oparin suggests that **life evolved from nonliving things**

Alexander Friedmann proposes a solution to Einstein's field equation that predicts that the **universe is expanding**

Gilbert Lewis and Merle Randall publish *Thermodynamics and the Free Energy of Chemical Substances*

Gilbert Lewis publishes *Valence and the Structure of Atoms and Molecules*, describing **electron-pair acid-base theory**

American mathematician George Birkhoff publishes *Relativity and Modern Physics*, describing **space-time**

September 10 Louis de Broglie suggests that electrons may have **wavelike properties**

October 6 Edwin Hubble observes that the **Andromeda nebula** is outside the Milky Way

February 14 IBM is founded in New York